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INVENTORY MANAGEMENT SYSTEM FOR JDN PHARMACY

A Capstone Proposal Presented to

Bachelor of Science in Information Systems

Carlos Hilado Memorial State of University-Fortune Towne Campus

In Partial Fulfillment of the Requirements

for the subject of the PROMAN 313

By

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Phase I

Initiation

Gap Analysis

JDN Pharmacy, located along Burgos Avenue in Bacolod City, currently relies on manual inventory tracking to manage its stock of medicines. Staff record incoming and outgoing supplies by hand, verify expiration dates manually, and monitor reorder levels without automated support. This traditional approach has led to several operational challenges that affect efficiency, accuracy, and customer service.

Inventory records are fragmented and dependent on staff diligence. Because of this, inconsistencies occur in stock counts, and decision-making is often delayed. When medicines are frequently dispensed, staff cannot quickly determine current stock levels or identify which items are nearing depletion. This increases the risk of stockouts, overstocking, or overlooking expired medicines.

Manual monitoring also makes it difficult to generate timely reports for management. Without a centralized system, pharmacy staff spend significant time preparing summaries of available stock, which slows down planning and reduces responsiveness to customer demand.

Handwritten records are vulnerable to human error and data loss. Once records are misplaced or damaged, there is no backup copy of inventory transactions. This undermines accountability and makes it difficult to track medicine demand or supplier performance.

The absence of a digital system prevents the pharmacy from scaling operations and adapting to modern customer expectations. Staff productivity is reduced because more time is spent on manual record-keeping instead of focusing on customer service.



Because of these problems—fragmented records, delayed reporting, risk of data loss, and inefficiency in monitoring stock levels—there is a clear need to develop a web-based Inventory Management System for JDN Pharmacy.

Project Title

Inventory Management System for JDN Pharmacy

Scope and Delimitation

Scope

This study focuses on developing a web-based Inventory Management System for JDN Pharmacy. The system will allow staff to register medicines, monitor stock levels, track expiry dates, record dispensing transactions, and generate basic reports. It will provide a centralized digital repository accessible across devices, designed to improve accuracy, efficiency, and reliability in managing pharmacy inventory. The system also includes role-based access control that distinguishes between administrator and staff permissions, as well as modules for managing drug categories, supplier information, and system user accounts. Automated alerts for low-stock and expiring medicines will be provided to support timely decision-making by pharmacy management.

Delimitation

The study is limited to inventory tracking for a single branch of JDN Pharmacy. It excludes advanced features such as automated purchase order generation, integration with point-of-sale systems, or native mobile application development. Financial accounting and billing modules are also not within scope.



General Objective

The primary goal of this research is to develop a web-based Inventory Management System that will streamline pharmaceutical stock management and improve operational efficiency at JDN Pharmacy, reducing manual workload, minimizing human error, and supporting informed decision-making through accurate and real-time inventory data.

Specific Objectives

1. To develop a centralized web-based repository that records and monitors medicine stock levels, including expiry dates and reorder points, ensuring that inventory data remains accurate, consistent, and accessible to authorized users at all times.
2. To design a complete medicine management module that supports the addition, updating, and removal of drug records—including details such as name, generic equivalent, SKU, category, dosage form, supplier, unit cost, and controlled substance flags—ensuring that inventory data remains accurate and up to date.
3. To implement an automated expiry monitoring and alert system that classifies drugs by urgency status and notifies staff of expired, critically expiring, and low-stock medicines without requiring manual checking, reducing the risk of dispensing expired or unavailable medicines.
4. To build a dispense logging module that records every drug dispensing transaction with details such as drug name, quantity dispensed, dispensing staff, and timestamp, and



automatically deducts the dispensed quantity from the live inventory to maintain accurate stock levels.

5. To evaluate the system's usability and effectiveness through structured User Acceptance Testing (UAT) conducted with actual pharmacy staff, using the gathered feedback to validate that the system meets operational needs and to make necessary improvements prior to final deployment.

Schematic/Conceptual Framework

Input–Process–Output–Outcome (IPOO) framework of the proposed Pharmacy Inventory system, showing how stakeholder needs and existing operational practices are processed through system development phases.

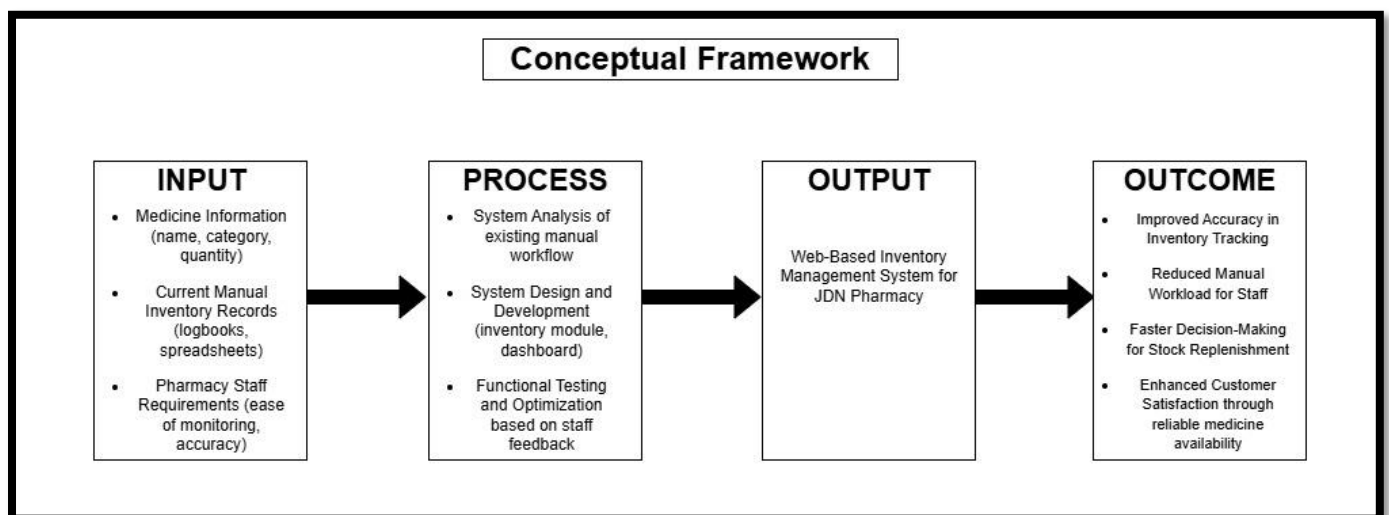


Figure I. Conceptual Framework